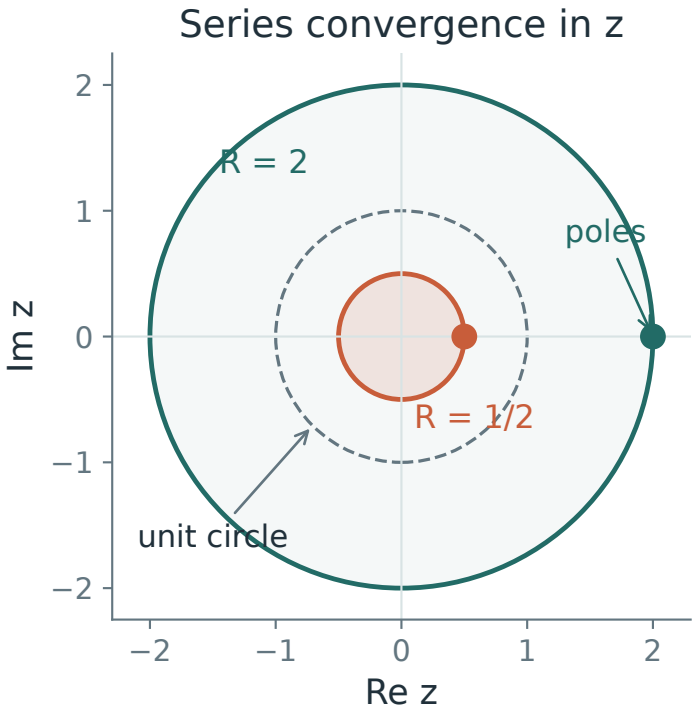
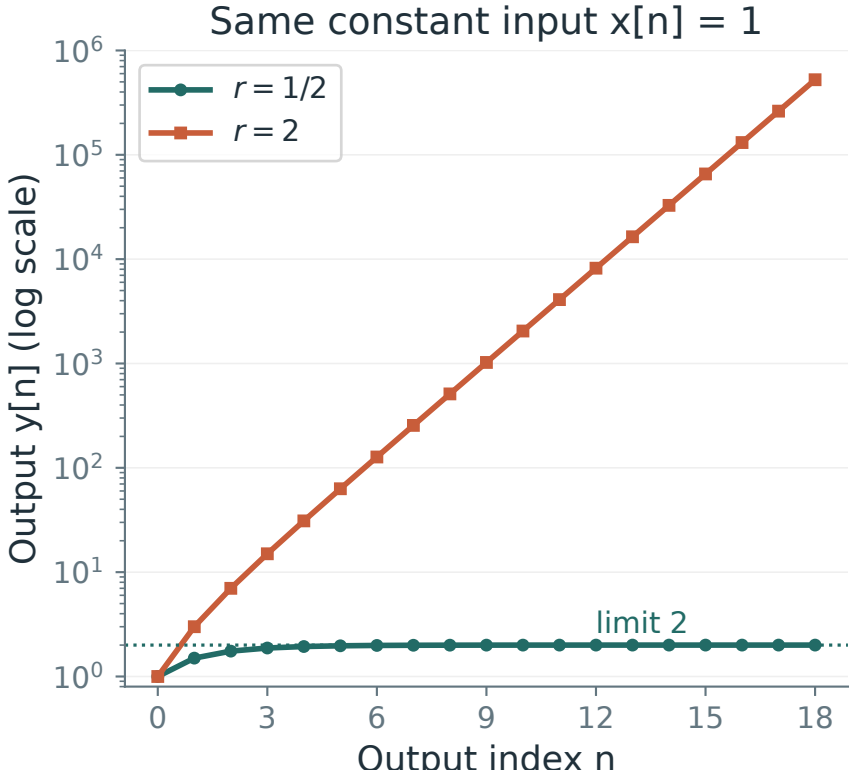

Series convergence in z
R = 2
R = 1/2
poles
unit circle
Re z
Im z
2
1
0
−1
−2
−2
−1
0
1
2


Same constant input x[n] = 1
r = 1/2
r = 2
limit 2
Output y[n] (log scale)
Output index n
10^6
10^5
10^4
10^3
10^2
10^1
10^0
0
3
6
9
12
15
18